QLORA: Efficient Finetuning of Quantized LLMs

Tim Dettmers; Artidoro Pagnoni; Ari Holtzman; Luke Zettlemoyer

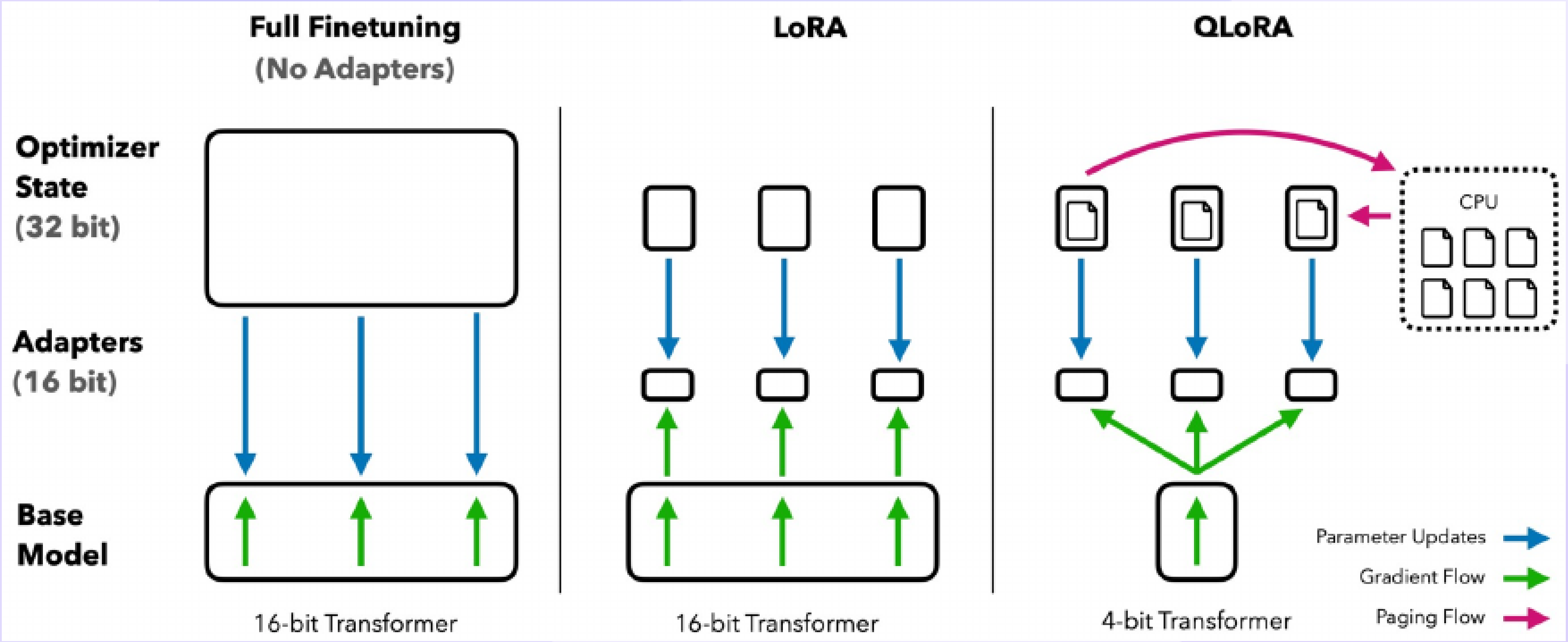
University of Washington

01 QLORA: Title and Authors

- Tim Dettmers ∗; Artidoro Pagnoni ∗; Ari Holtzman; Luke Zettlemoyer
- University of Washington
  - {dettmers, artidoro, ahai, lsz}@cs.washington.edu

- QLORA: Efficient finetuning of quantized large language models

02 QLORA: What, How, Outcomes

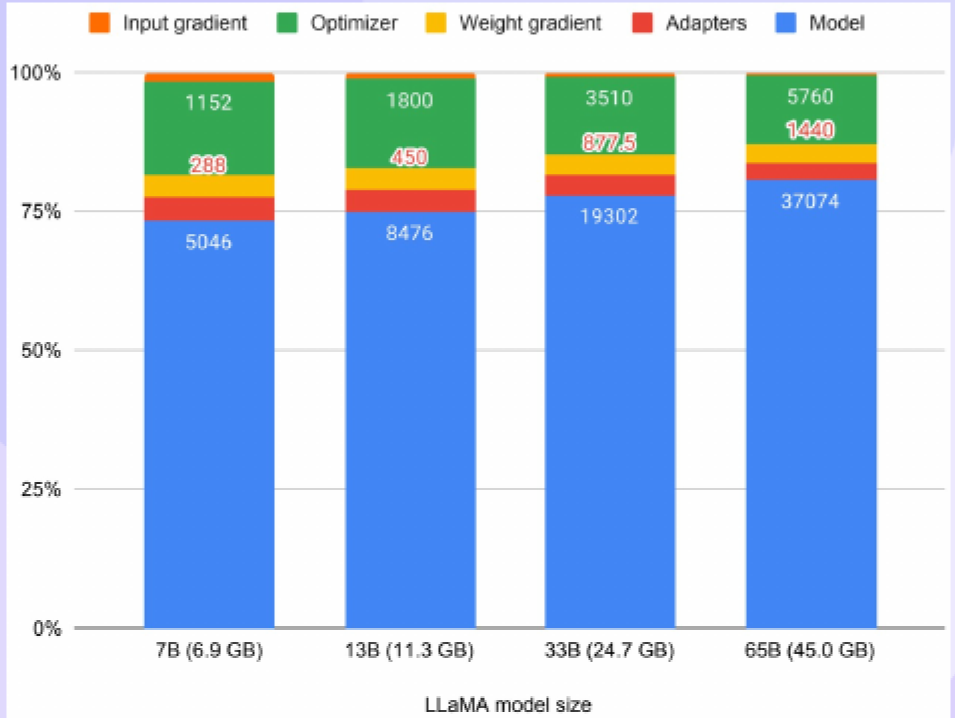


- QLORA backpropagates through a frozen 4-bit base into LoRA adapters with full 16-bit parity
- NF4 quantization, double quantization, and paged optimizers enable stable finetuning
- Guanaco achieves near-ChatGPT quality and scales to 65B on one 48 GB GPU

03 Memory Problem and Approach

- Sixteen-bit finetuning of 65B needs more than 780 GB and risks out-of-memory spikes
- QLORA reduces working memory to under 48 GB using 4-bit NF4 and paged optimizers
- Training updates apply only to LoRA while compute runs in BF16 after dequantization

04 GPU Memory and Tradeoffs



- 24 GB: 7B/13B; 48 GB: 65B with QLORA
- Memory driven by params, optimizer states, activations
- Grad accumulation and paged optimizers prevent spikes

05 Efficiency and Performance

- QLORA cuts memory from more than 780 GB to under 48 GB with parity to 16-bit finetuning
- Guanaco 65B reaches near-ChatGPT performance within roughly one day
- NF4, double quantization, and paged optimizers drive **efficiency** and **stability**

06 Motivation Recap

- Sixteen-bit finetuning is memory prohibitive for 65B and suffers peak usage spikes
- Naive low-bit training destabilizes gradients and leaves memory as the bottleneck
- QLORA freezes 4-bit weights and trains LoRA with **stable** and **efficient** updates

07 QLORA in Context

- Inference quantization methods improve forward fidelity but rarely address training stability
- Backpropagation through quantized weights is explored at scale and aligns with QLORA design
- Parameter-efficient finetuning options exist, while LoRA offers **robust** and **simple** integration

[66]; [14]; [44]; [69]; [13]; [71]; [47]; [18]; [65]; [28]; [48]; [33]; [34]; [1]; [37]; [27]; [70]; [54]; [23]; [40]...

Why Prior Quantization Fails

- Block-wise quantization improves bin use but leaves gradient flow unstable during training
- Outliers and coarse bins distort updates and increase training variance
- Training-aware design is required to preserve **stability** and **accuracy**

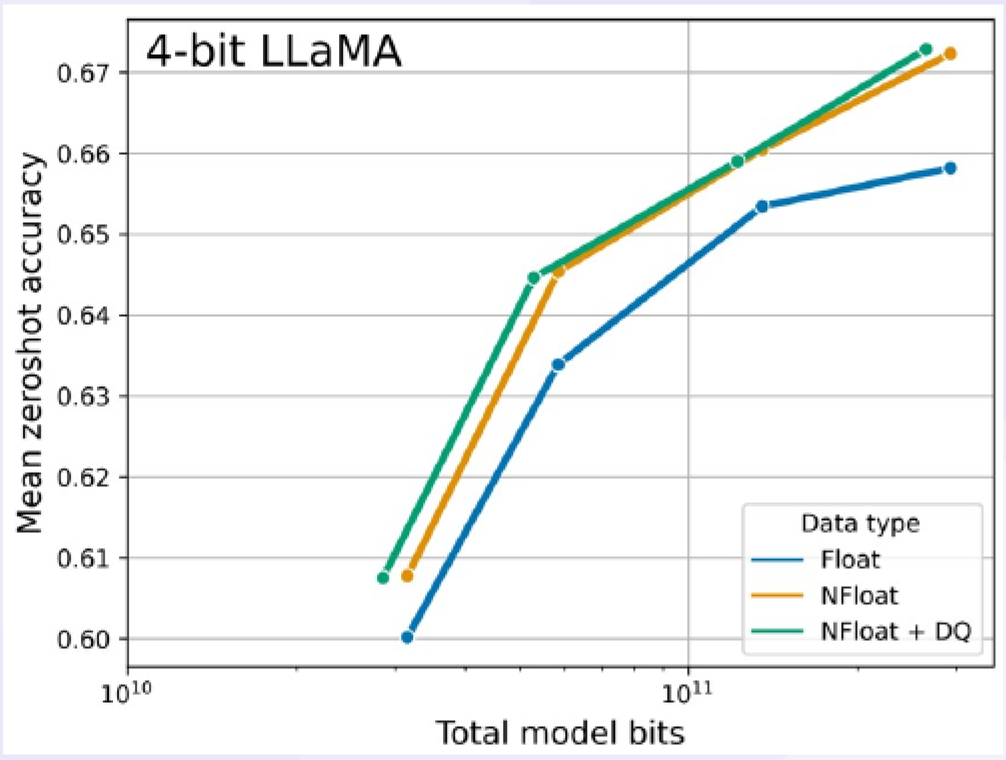
Adapters Don't Solve Memory

- LoRA parameter counts are small while activations and gradients dominate memory
- Gradient checkpointing is essential and enables multiple adapters with low overhead
- Reducing adapter size yields limited savings compared to **activation memory**

Study Scope and Themes

- More than one thousand finetunes span 80M to 65B across models and datasets
- QLORA matches strong 16-bit baselines and powers Guanaco chatbots
- Data quality outweighs size for **instruction following** performance

NF**4** Quantization



- NF4 uses fixed quantiles for zero-centered weights and provides an exact zero level
- Weights are stored in 4-bit and dequantized to BF16 for computation
- NF4 outperforms other 4-bit types and is robust to outliers

12 DQ + Paged Optimizers

- Double quantization compresses scales, reducing bits per parameter
- Paged optimizers use unified memory to avoid OOM
- Lower overhead preserves quality, enables larger models

13 Training Recipe

- Weights are stored in NF4 with dequantization of scales and computation in BF16
- Backpropagation updates only LoRA parameters while base weights remain frozen
- One storage type and one compute type simplify **engineering** and **deployment**

14 Guanaco and MMLU

- Guanaco 65B: near-ChatGPT on Vicuna; strong Elo
- FLAN v2 improves MMLU, aligns poorly with chat metrics
- Small high-quality data beats larger mixed sets

15 NF**4** Evidence and Parity

- NF4 beats FP4/Int4 on accuracy and perplexity.
- Four-bit QLORA matches strong 16-bit baselines.
- Memory efficiency lets larger 4-bit models win.

16 Comparison: **16-**bit Baseline

- GLUE: QLORA matches 16-bit (≈88.6–88.8)
- MMLU: NF4 ≈ 16-bit; FP4 ~1 point behind
- Memory: 780 GB vs <48 GB (65B finetuning)
- Guanaco 65B trains ≈24 h on one GPU

17 Protocols and Setups

- Benchmarks cover GLUE, TKInstruct, and LLaMA from 7B to 65B for generative tasks
- Training uses NF4 with double quantization and paged optimizers across all linear layers
- Evaluation includes Vicuna and Open Assistant with human and GPT-4 pairwise judgments

[GLUE]; [TKInstruct]; [Vicuna]; [OpenAssistant]; [GPT-4]

18 Ablations That Matter

- Placing adapters in all linear layers best preserves parity with full finetuning
- LoRA rank between eight and thirty-two shows similar quality while very small ranks underfit
- NF4 block size sixty-four balances **accuracy** and **overhead** effectively

19 Qualitative Biases and Limits

- Models show strong recall on common facts and overconfidence on obscure items
- Automated judges exhibit order bias and self-preference in pairwise scoring
- Human agreement is moderate and varies with system strength and task type

Limits and Open Space

- Verification at 33B and 65B remains resource intensive across broader tasks
- Coverage omits several benchmarks and limits generalization claims
- Unexplored precisions and adapters suggest future headroom for efficiency

[BigBench]; [RAFT]; [HELM]

Impacts and Responsible Use

- Democratized finetuning improves access and supports transparent research
- Lower barriers can amplify misuse without safeguards and oversight
- Mitigations include model cards, provenance, filters, and targeted red-teaming

[Model Cards]

Conclusion and Resources

- QLORA enables 33B on consumer hardware and 65B on a single 48 GB GPU with parity
- Released models and code support rapid adoption and reproducibility
- On-device finetuning of 7B-class models becomes feasible for private applications

[Code Release]

Thank You

Questions & Discussion